

Table 1: Clean test MSE with additive Gaussian noise $\sigma = 0.1$; mean $\pm$ sample standard deviation over 2 seeds. Brackets give $100 \text{MSE}_{\text{method}}/\text{MSE}_{\text{formula}}$ using the respective seed means. Inputs are uniform on $[-1, 1]^{100}$, with $N_{\text{train}} = 1000$ and $N_{\text{test}} = 2000$. The formula selects the averaging radius β by 5-fold CV over 30 predefined candidates using noisy training labels, then recomputes all averages using the full training set. The formula uses fixed $L = 1$; SN and orthogonal networks have bound 1. All MLPs use 3 hidden ReLU layers and 1001 full-batch Adam updates with learning rate 0.001, without early stopping.

(a) Hard

Target	Formula	ReLU MLP	SN MLP	Orthogonal MLP	Constant
64 signed Voronoi tents	$(4.846 \pm 0.002) \times 10^{-2}$ [100.0%]	$(5.898 \pm 0.042) \times 10^{-2}$ [121.7%]	$(5.641 \pm 0.037) \times 10^{-2}$ [116.4%]	$(5.263 \pm 0.042) \times 10^{-2}$ [108.6%]	$(4.847 \pm 0.003) \times 10^{-2}$ [100.0%]

(b) Tied

Target	Formula	ReLU MLP	SN MLP	Orthogonal MLP	Constant
Folding: 1 layer	$(1.201 \pm 0.038) \times 10^{-1}$ [100.0%]	$(1.424 \pm 0.131) \times 10^{-1}$ [118.5%]	$(1.244 \pm 0.063) \times 10^{-1}$ [103.6%]	$(1.258 \pm 0.051) \times 10^{-1}$ [104.7%]	$(1.200 \pm 0.036) \times 10^{-1}$ [99.9%]
Folding: 4 layers	$(4.225 \pm 0.278) \times 10^{-2}$ [100.0%]	$(5.436 \pm 0.261) \times 10^{-2}$ [128.7%]	$(5.198 \pm 0.220) \times 10^{-2}$ [123.0%]	$(4.903 \pm 0.264) \times 10^{-2}$ [116.0%]	$(4.219 \pm 0.269) \times 10^{-2}$ [99.9%]

Notes. Hard and Tied are target-group labels, not conclusions about noisy results. Signed Voronoi tents replace the earlier disjoint-ball bumps to obtain nonzero uniform samples in 100 dimensions; each target is 1-Lipschitz. Stored scalars: formula 101,001; each MLP 100,801 (ratio 0.9980, width 200). The constant predicts the noisy training-label mean. Only the formula’s radius is cross-validated; the neural training schedule is fixed.

Table 2: Clean test MSE with additive Gaussian noise $\sigma = 0.2$; mean $\pm$ sample standard deviation over 2 seeds. Brackets give $100 \text{MSE}_{\text{method}}/\text{MSE}_{\text{formula}}$ using the respective seed means. Inputs are uniform on $[-1, 1]^{100}$, with $N_{\text{train}} = 1000$ and $N_{\text{test}} = 2000$. The formula selects the averaging radius β by 5-fold CV over 30 predefined candidates using noisy training labels, then recomputes all averages using the full training set. The formula uses fixed $L = 1$; SN and orthogonal networks have bound 1. All MLPs use 3 hidden ReLU layers and 1001 full-batch Adam updates with learning rate 0.001, without early stopping.

(a) Hard

Target	Formula	ReLU MLP	SN MLP	Orthogonal MLP	Constant
64 signed Voronoi tents	$(4.843 \pm 0.009) \times 10^{-2}$ [100.0%]	$(6.945 \pm 0.265) \times 10^{-2}$ [143.4%]	$(6.257 \pm 0.016) \times 10^{-2}$ [129.2%]	$(5.758 \pm 0.018) \times 10^{-2}$ [118.9%]	$(4.844 \pm 0.009) \times 10^{-2}$ [100.0%]

(b) Tied

Target	Formula	ReLU MLP	SN MLP	Orthogonal MLP	Constant
Folding: 1 layer	$(1.202 \pm 0.039) \times 10^{-1}$ [100.0%]	$(1.541 \pm 0.155) \times 10^{-1}$ [128.2%]	$(1.295 \pm 0.072) \times 10^{-1}$ [107.8%]	$(1.278 \pm 0.055) \times 10^{-1}$ [106.4%]	$(1.200 \pm 0.036) \times 10^{-1}$ [99.8%]
Folding: 4 layers	$(4.229 \pm 0.278) \times 10^{-2}$ [100.0%]	$(6.687 \pm 0.439) \times 10^{-2}$ [158.1%]	$(5.894 \pm 0.230) \times 10^{-2}$ [139.4%]	$(5.301 \pm 0.362) \times 10^{-2}$ [125.3%]	$(4.221 \pm 0.265) \times 10^{-2}$ [99.8%]

Notes. Hard and Tied are target-group labels, not conclusions about noisy results. Signed Voronoi tents replace the earlier disjoint-ball bumps to obtain nonzero uniform samples in 100 dimensions; each target is 1-Lipschitz. Stored scalars: formula 101,001; each MLP 100,801 (ratio 0.9980, width 200). The constant predicts the noisy training-label mean. Only the formula’s radius is cross-validated; the neural training schedule is fixed.

Table 3: Clean test MSE with additive Gaussian noise $\sigma = 0.8$; mean $\pm$ sample standard deviation over 2 seeds. Brackets give $100 \text{MSE}_{\text{method}}/\text{MSE}_{\text{formula}}$ using the respective seed means. Inputs are uniform on $[-1, 1]^{100}$, with $N_{\text{train}} = 1000$ and $N_{\text{test}} = 2000$. The formula selects the averaging radius β by 5-fold CV over 30 predefined candidates using noisy training labels, then recomputes all averages using the full training set. The formula uses fixed $L = 1$; SN and orthogonal networks have bound 1. All MLPs use 3 hidden ReLU layers and 1001 full-batch Adam updates with learning rate 0.001, without early stopping.

(a) Hard

Target	Formula	ReLU MLP	SN MLP	Orthogonal MLP	Constant
64 signed Voronoi tents	$(4.951 \pm 0.005) \times 10^{-2}$ [100.0%]	$(2.751 \pm 0.003) \times 10^{-1}$ [555.7%]	$(1.134 \pm 0.021) \times 10^{-1}$ [229.1%]	$(1.119 \pm 0.037) \times 10^{-1}$ [225.9%]	$(4.909 \pm 0.066) \times 10^{-2}$ [99.2%]

(b) Tied

Target	Formula	ReLU MLP	SN MLP	Orthogonal MLP	Constant
Folding: 1 layer	$(1.206 \pm 0.045) \times 10^{-1}$ [100.0%]	$(3.501 \pm 0.046) \times 10^{-1}$ [290.4%]	$(1.740 \pm 0.074) \times 10^{-1}$ [144.3%]	$(1.716 \pm 0.125) \times 10^{-1}$ [142.3%]	$(1.206 \pm 0.044) \times 10^{-1}$ [100.0%]
Folding: 4 layers	$(4.352 \pm 0.187) \times 10^{-2}$ [100.0%]	$(2.963 \pm 0.090) \times 10^{-1}$ [680.8%]	$(1.089 \pm 0.032) \times 10^{-1}$ [250.3%]	$(1.075 \pm 0.003) \times 10^{-1}$ [247.0%]	$(4.313 \pm 0.132) \times 10^{-2}$ [99.1%]

Notes. Hard and Tied are target-group labels, not conclusions about noisy results. Signed Voronoi tents replace the earlier disjoint-ball bumps to obtain nonzero uniform samples in 100 dimensions; each target is 1-Lipschitz. Stored scalars: formula 101,001; each MLP 100,801 (ratio 0.9980, width 200). The constant predicts the noisy training-label mean. Only the formula’s radius is cross-validated; the neural training schedule is fixed.